Introduction I

- Some features of permutations and combinations that are not directly related to counting.
- Some ordering schemes for permutations and combinations, as well as algorithms for carrying them out.
- The idea of a relation on a set and two important instances, those of partial order and equivalence relation.



Generating Permutations I

- *Stirling's formula* (Feller, 1967)

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

- Algorithm for generating all the permutations of $\{1, 2, \cdots, n\}$ (Johnson, 1963; Trotter, 1962; Gardner, 1974), is based on the following observation

  If the integer n is deleted from a permutation of $\{1, 2, \cdots, n\}$, the result is a permutation $\{1, 2, \cdots, n-1\}$.

Example

$n = 5$

$$\begin{array}{ccccc} 5, & 3, & 4, & 1, & 2 \\ 3, & 5, & 4, & 1, & 2 \\ 3, & 4, & 5, & 1, & 2 \\ 3, & 4, & 1, & 5, & 2 \\ 3, & 4, & 1, & 2, & 5 \end{array}$$

Generating Permutations II

- Given a list of the $(n-1)!$ permutations of $\{1, 2, \cdots, n-1\}$ we can obtain a list of the $n!$ permutations of $\{1, 2, \cdots, n\}$ by systematically inserting n into each permutation of $\{1, 2, \cdots, n-1\}$ in all possible ways.

Example

$n = 2$

	1	2
2	1	

$n = 3$

	1	2	3
	1	3	2
3	1	2	
3	2	1	
	2	3	1
	2	1	3

Generating Permutations III

Example

$n = 4$

			4	4							4	4							4	4			
1	1	1	1	1	1	1	1	3	3	3	3	3	3	3	3	2	2	2	2	2	2	2	2
		4			4					4			4					4			4		
2	2	2	2	3	3	3	3	1	1	1	1	2	2	2	2	3	3	3	3	1	1	1	1
	4					4			4					4			4					4	
3	3	3	3	2	2	2	2	2	2	2	2	1	1	1	1	1	1	1	1	3	3	3	3
4							4	4							4	4							4

- Give an integer k, we assign a *direction* to it by writing an arrow above it pointing to the left or to the right: $\overleftarrow{k}$ or $\overrightarrow{k}$.
- Consider a permutation of $\{1, 2, \ldots, n\}$ in which each of the integers is given a direction. The integer k is called mobile if its arrow points to a smaller integer adjacent to it.



Generating Permutations IV

- It follows that the integer 1 can never be mobile. The integer n is mobile, except that n is the first (last) integer and its arrow points to the left (right): $\overleftarrow{n} \cdots$ $(\cdots \overrightarrow{n})$.
- Algorithm for generating the permutations of $\{1, 2, \ldots, n\}$
  Begin with $\overleftarrow{1}\ \overleftarrow{2} \cdots \overleftarrow{n}$. While there exists a mobile integer, do the following:
  1. Find the largest mobile integer m.
  2. Switch m and the adjacent integer its arrow points to.
  3. Switch the direction of all the arrows above integers p with $p > m$.



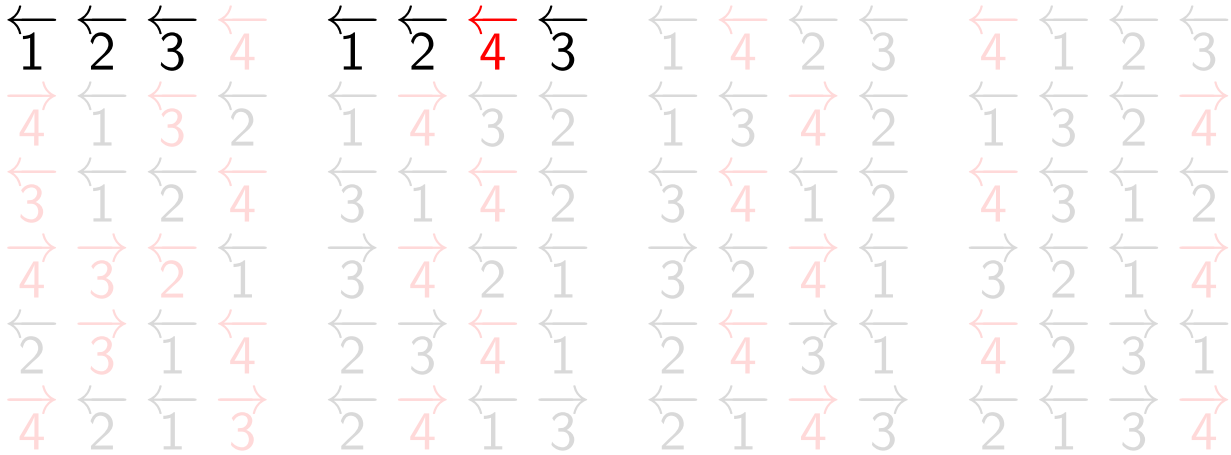
Generating Permutations V

Algorithm illustration for $n = 4$



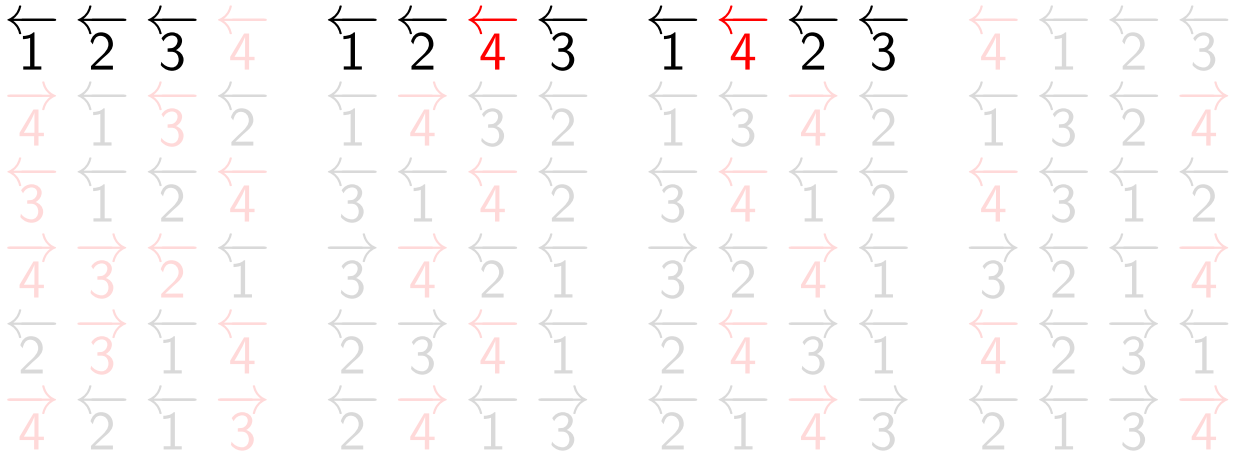
Generating Permutations V

Algorithm illustration for $n = 4$



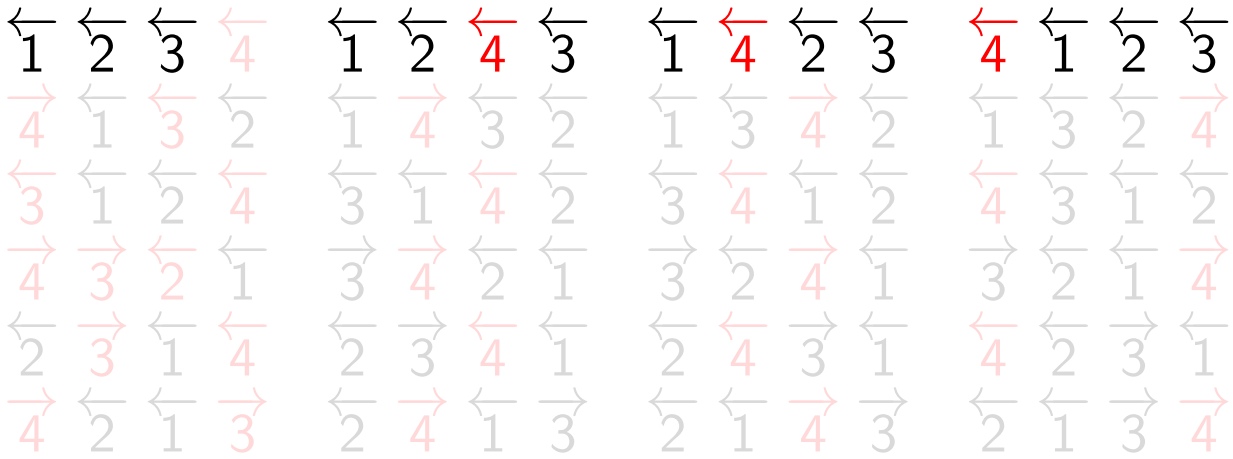
Generating Permutations V

Algorithm illustration for $n = 4$



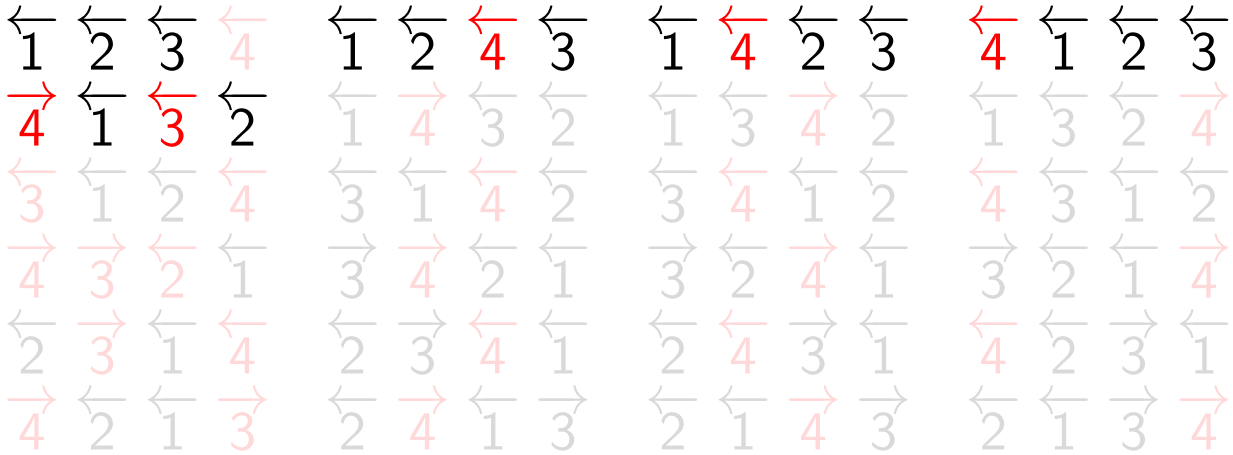
Generating Permutations V

Algorithm illustration for $n = 4$



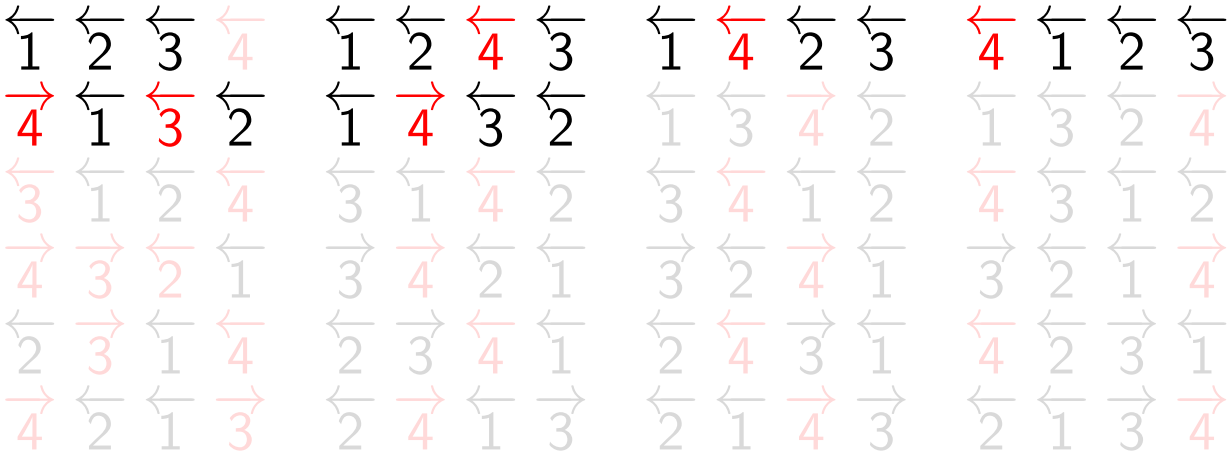
Generating Permutations V

Algorithm illustration for $n = 4$



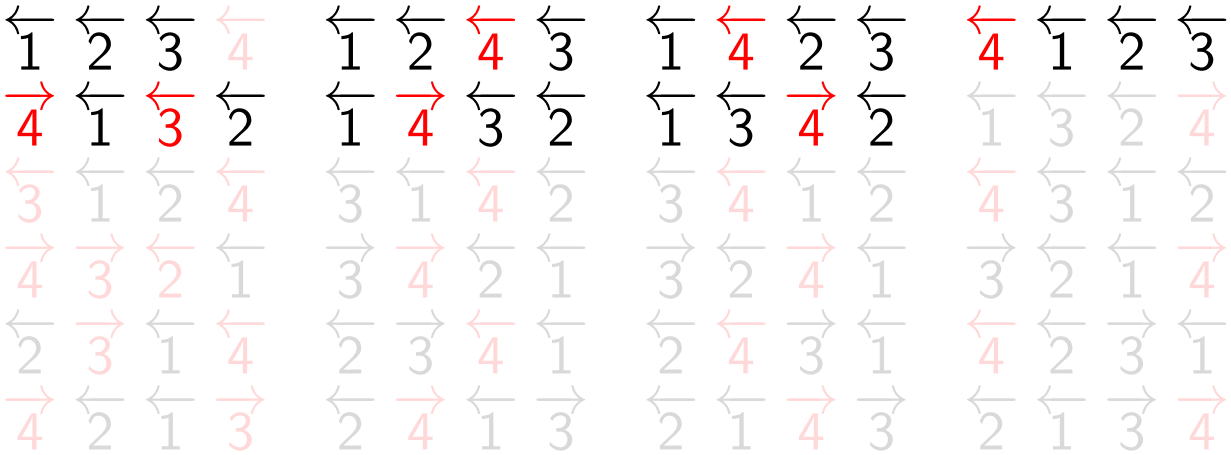
Generating Permutations V

Algorithm illustration for $n = 4$



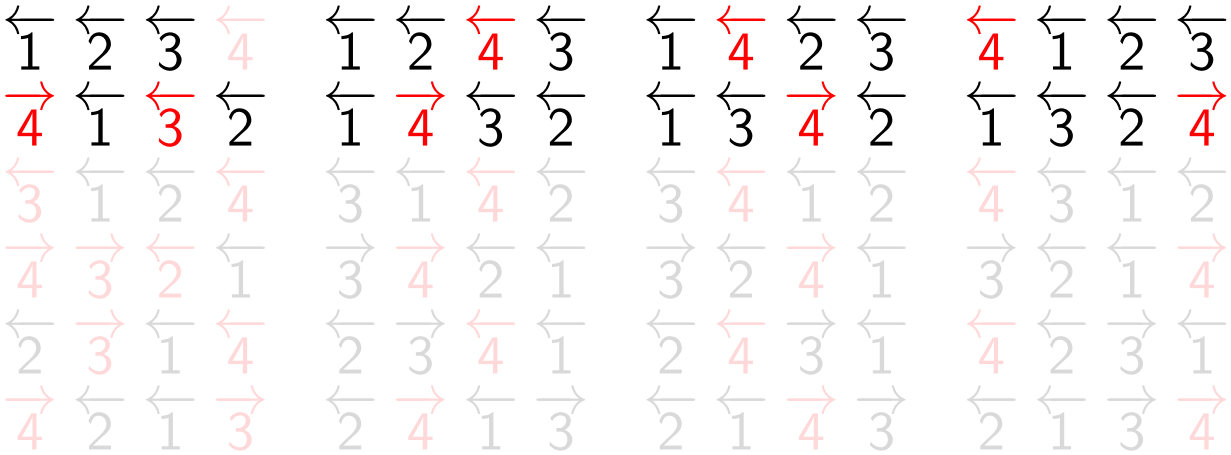
Generating Permutations V

Algorithm illustration for $n = 4$



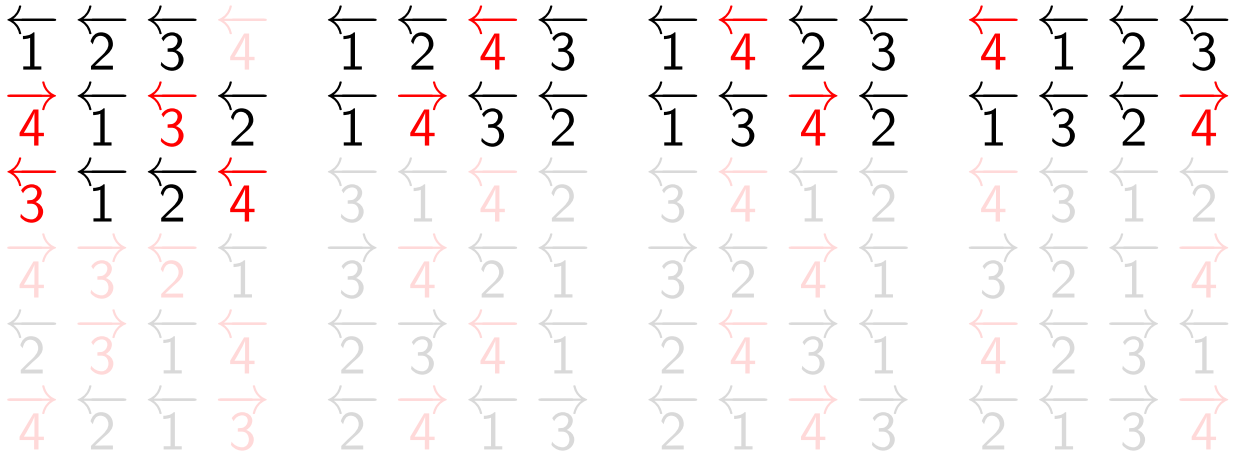
Generating Permutations V

Algorithm illustration for $n = 4$



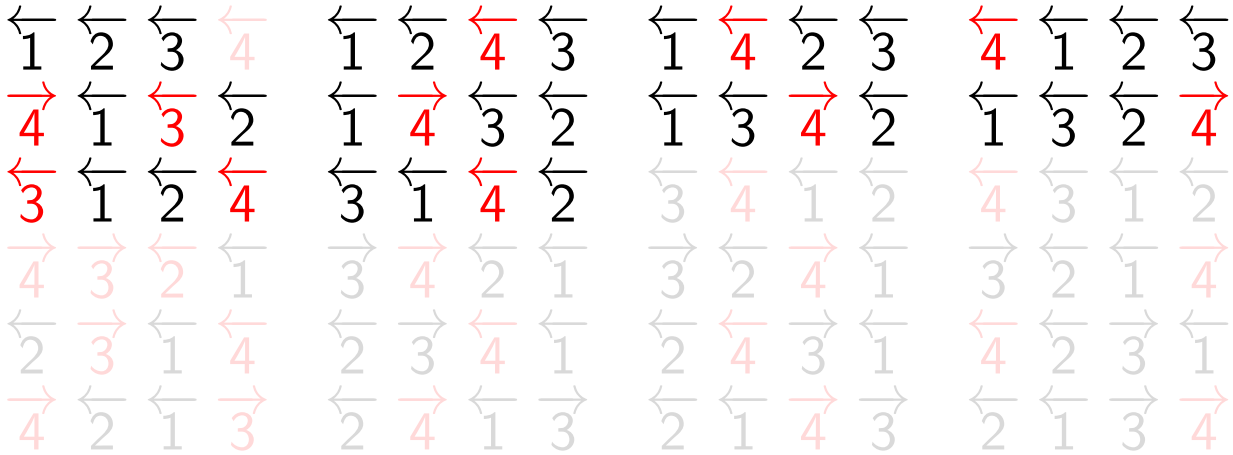
Generating Permutations V

Algorithm illustration for $n = 4$



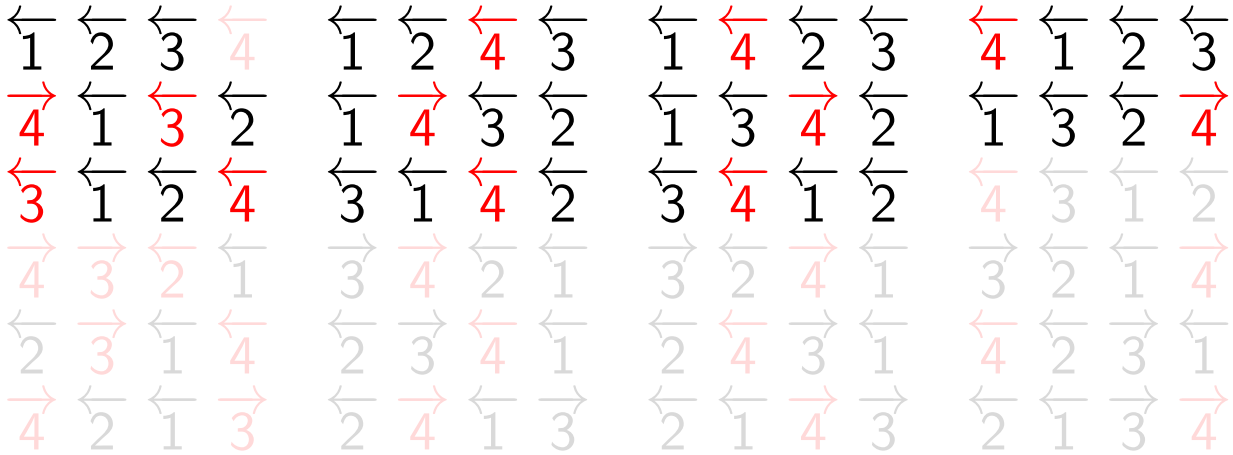
Generating Permutations V

Algorithm illustration for $n = 4$



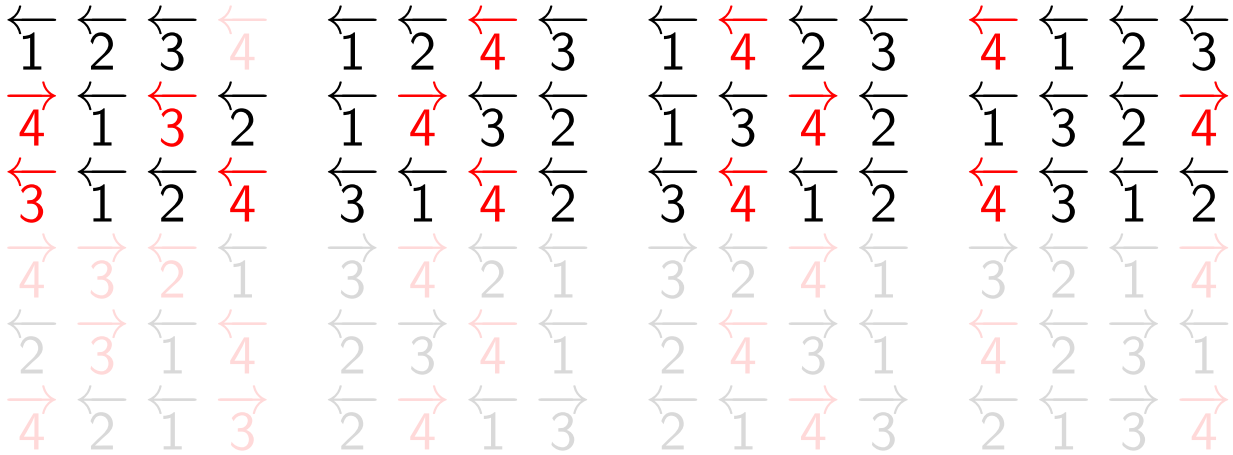
Generating Permutations V

Algorithm illustration for $n = 4$



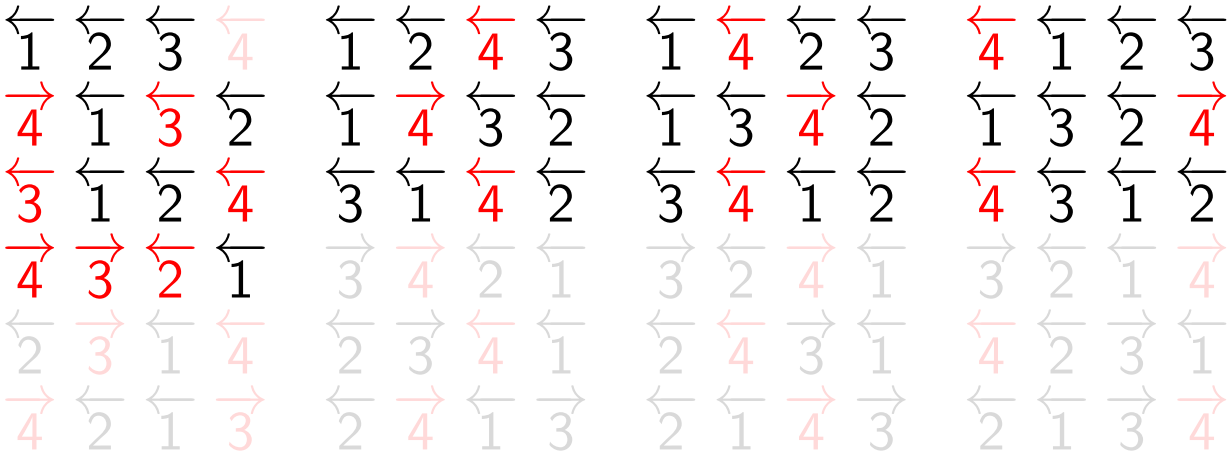
Generating Permutations V

Algorithm illustration for $n = 4$



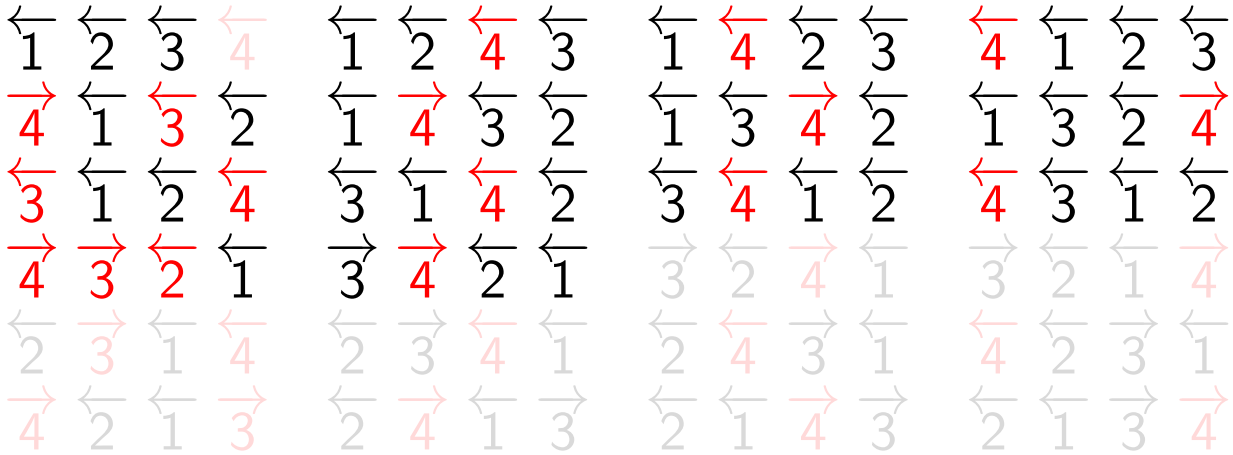
Generating Permutations V

Algorithm illustration for $n = 4$



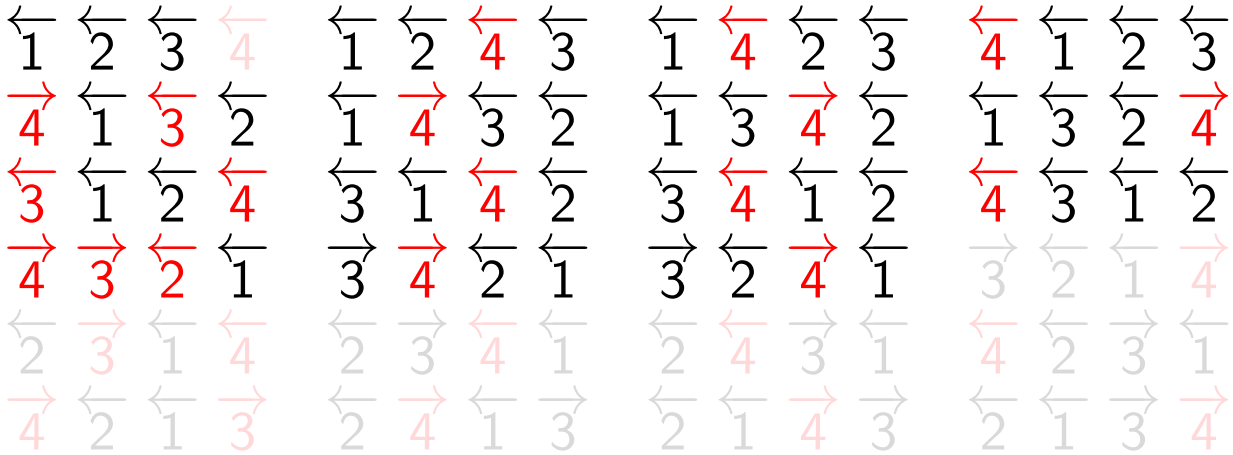
Generating Permutations V

Algorithm illustration for $n = 4$



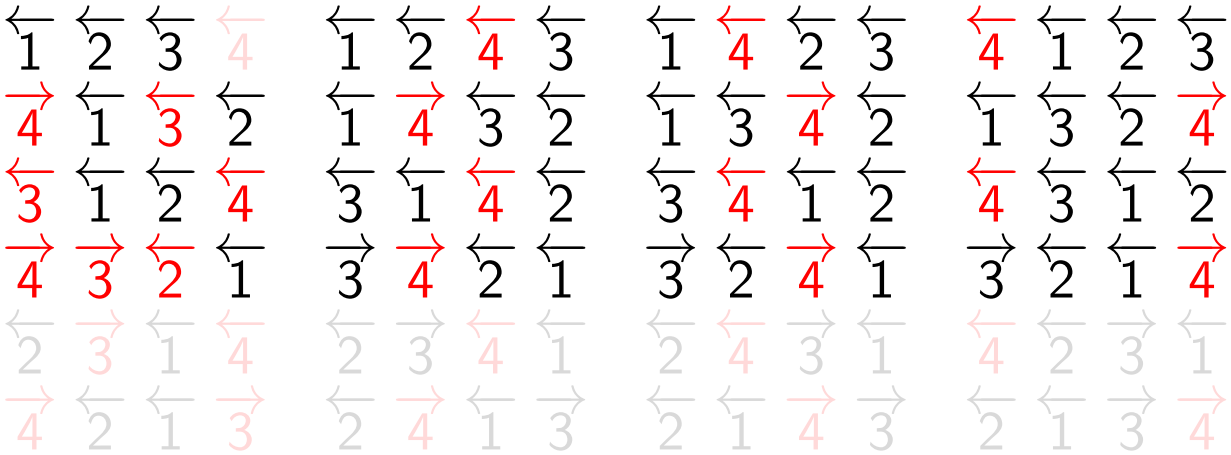
Generating Permutations V

Algorithm illustration for $n = 4$



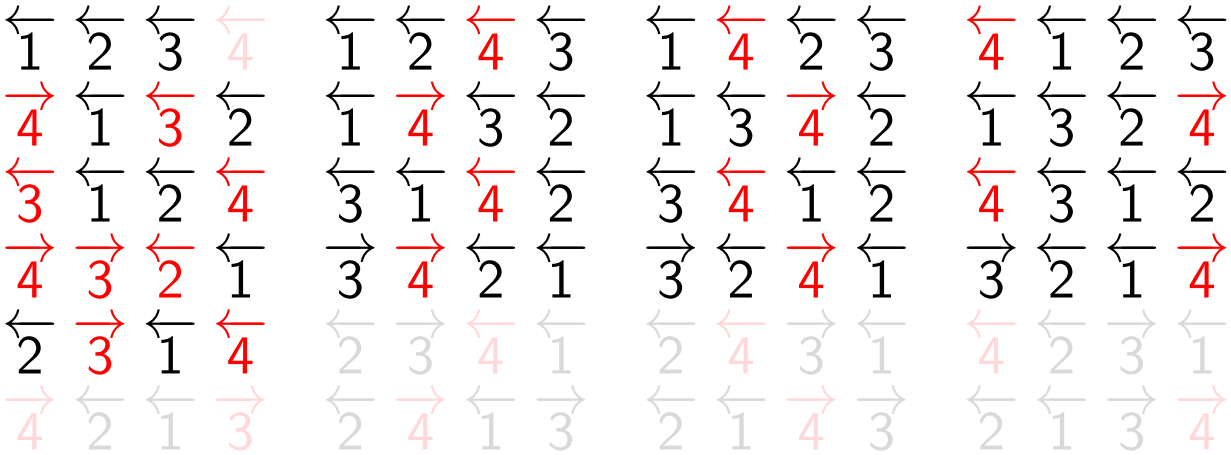
Generating Permutations V

Algorithm illustration for $n = 4$



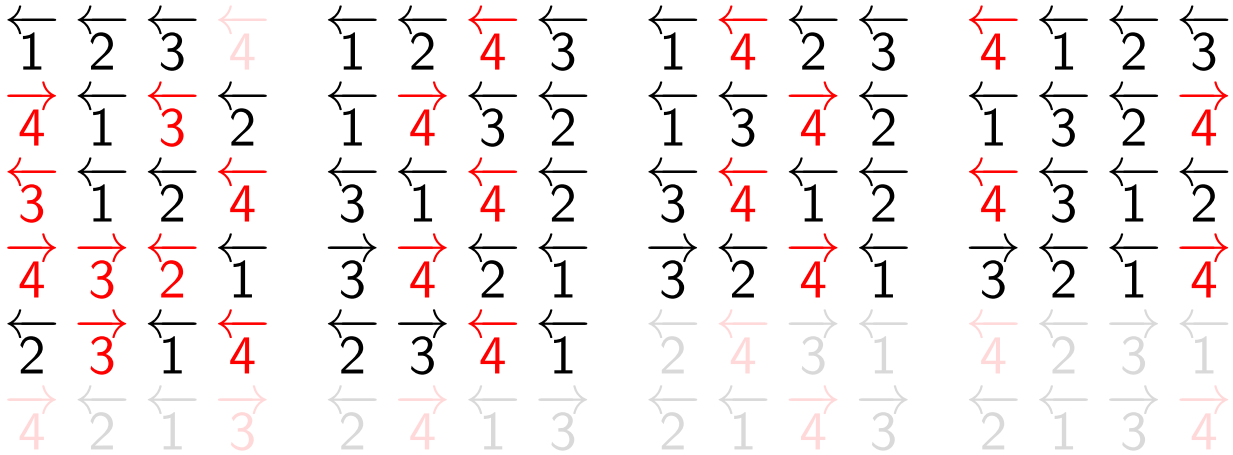
Generating Permutations V

Algorithm illustration for $n = 4$



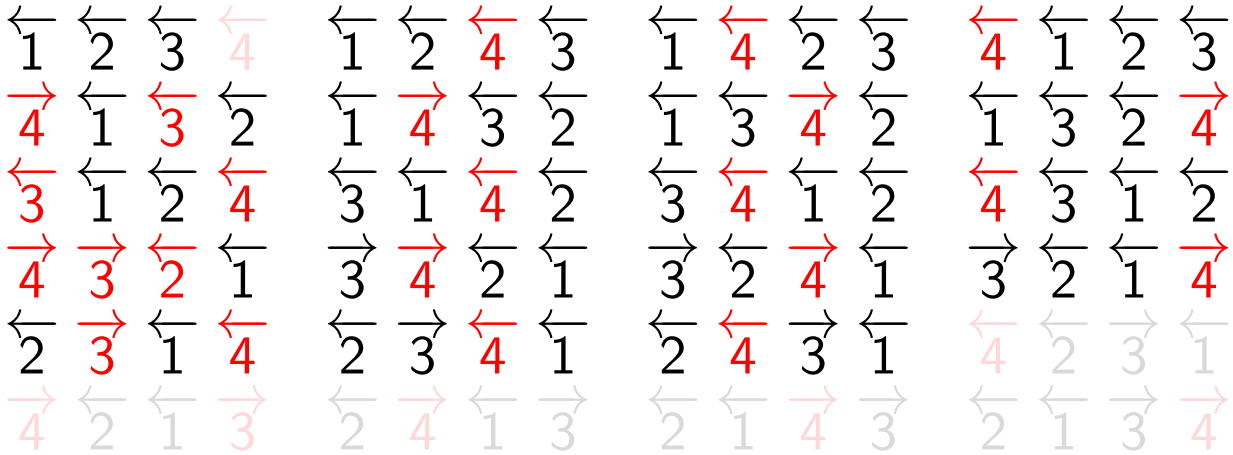
Generating Permutations V

Algorithm illustration for $n = 4$



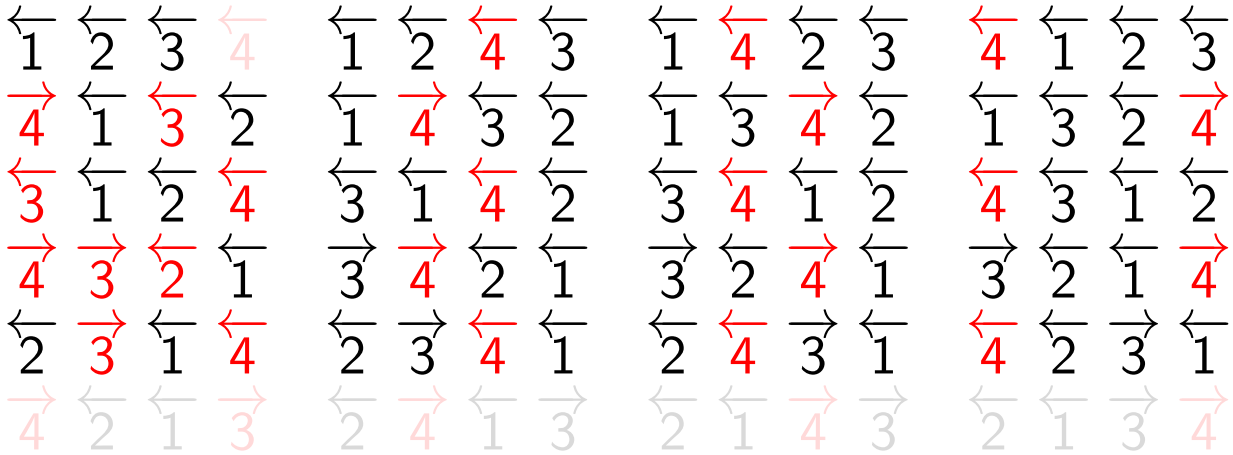
Generating Permutations V

Algorithm illustration for $n = 4$



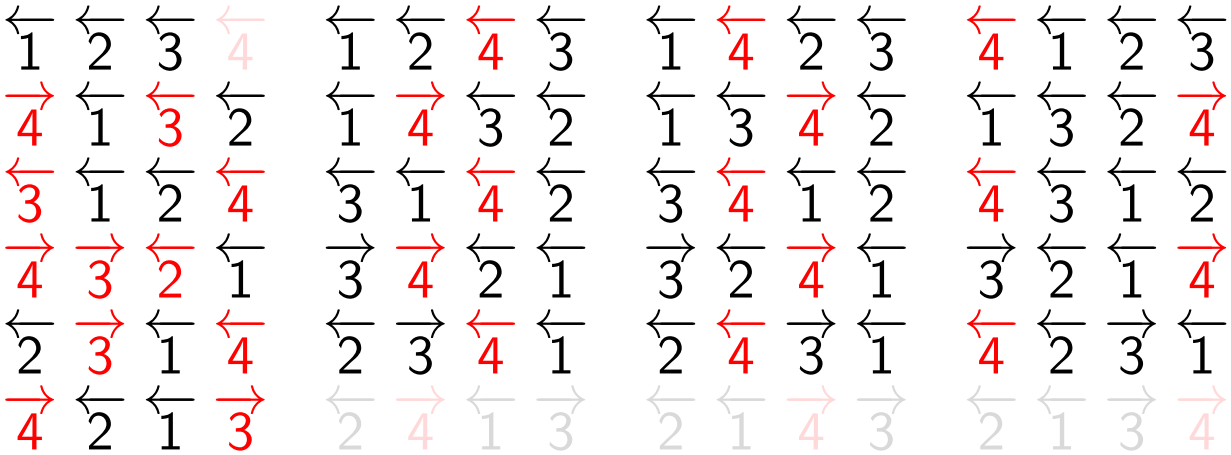
Generating Permutations V

Algorithm illustration for $n = 4$



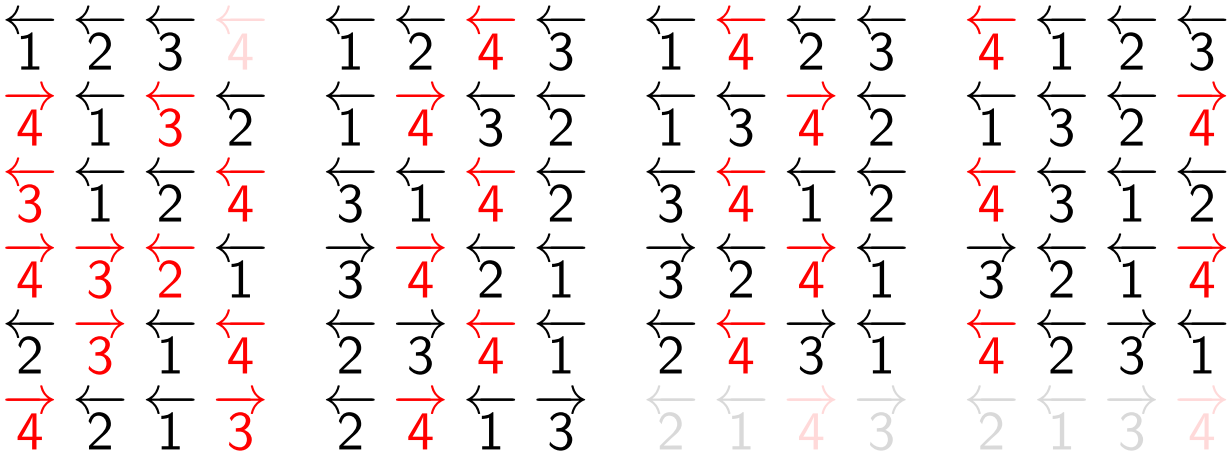
Generating Permutations V

Algorithm illustration for $n = 4$



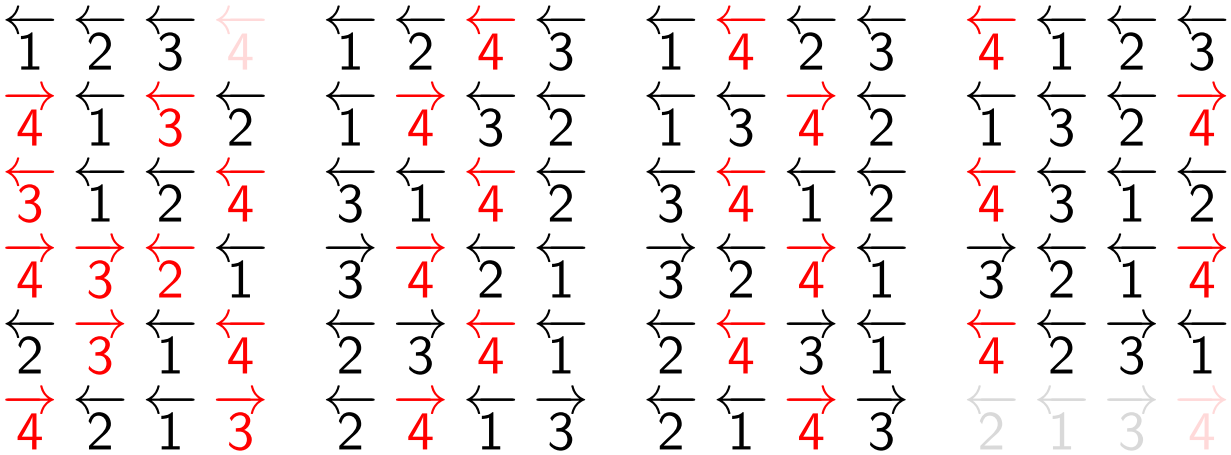
Generating Permutations V

Algorithm illustration for $n = 4$



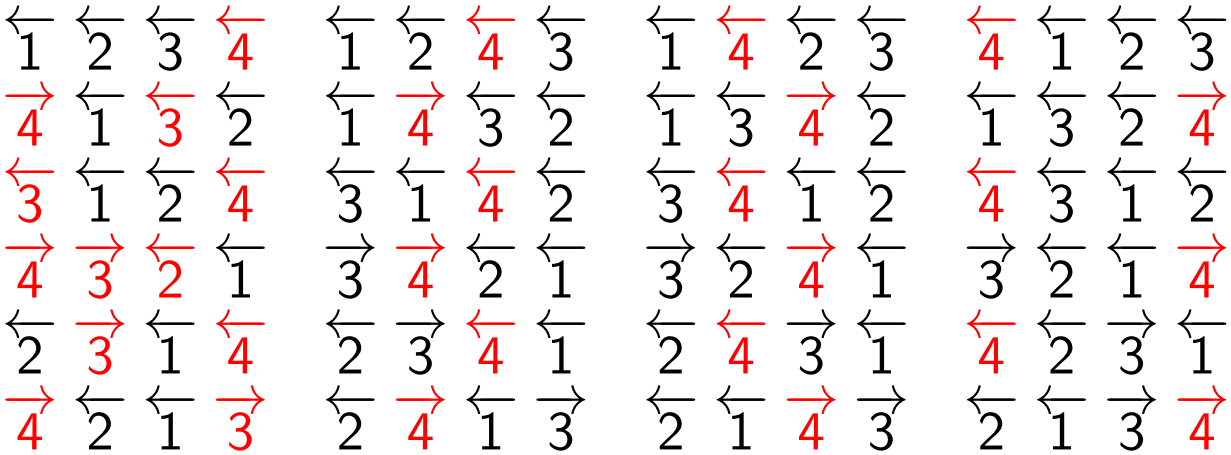
Generating Permutations V

Algorithm illustration for $n = 4$



Generating Permutations V

Algorithm illustration for $n = 4$



Inversions in Permutations I

- A method of describing a permutation by means of its inversions (Hall, 1963)
- Let $i_1 i_2 \dots i_n$ be a permutation of the set $\{1, 2, \dots, n\}$. The pair (i_k, i_l) is called an *inversion* if $k < l$ and $i_k > i_l$.

Example

The permutation 31524 has four inversions, namely (3,1), (3,2), (5,2), (5,4).

- The only permutation of $\{1, 2, \dots, n\}$ with no inversions is $12 \cdots n$.
- For a permutation $i_1 i_2 \cdots i_n$ we let a_j denote the number of inversions whose second component is j. In other words,

  a_j equals the number of integers that precede j in the permutation, but are greater than j; it measures how much j is out of order.



Inversions in Permutations II

- The sequence of numbers $a_1, a_2, \ldots, a_n$ is called the *inverse sequence* of the permutation $i_1 i_2 \cdots i_n$. The number $a_1 + a_2 + \cdots + a_n$ measures the *disorder* of a permutation.

Example

The inversion sequence of the permutation 31524 is

$$1, 2, 0, 1, 0.$$

- The inverse sequence of numbers $a_1, a_2, \ldots, a_n$ of the permutation $i_1 i_2 \ldots i_n$ satisfies the conditions

$$0 \leq a_1 \leq n-1,\ 0 \leq a_2 \leq n-2,\ \ldots,\ 0 \leq a_{n-1} \leq 1,\ a_n = 0.$$



Inversions in Permutations III

- Using the multiplication principle, The number of sequences of integers $b_1, b_2, \ldots, b_n$, with

$$0 \le b_1 \le n-1,\ 0 \le b_2 \le n-2,\ \ldots,\ 0 \le b_{n-1} \le 1,\ b_n = 0,$$

equals $n \times (n-1) \times \cdots \times 2 \times 1 = n!$.

Theorem

Let $b_1, b_2, \ldots, b_n$ be a sequence of integers satisfying

$$0 \le b_1 \le n-1,\ 0 \le b_2 \le n-2,\ \ldots,\ 0 \le b_{n-1} \le 1,\ b_n = 0.$$

Then there exists a unique permutation of $\{1, 2, \ldots, n\}$ whose inversion sequence is $b_1, b_2, \ldots, b_n$.

- Algorithm I

  n: Write down n.



Inversions in Permutations IV

$n-1$: Consider b_{n-1}. We are given that $0 \le b_{n-1} \le 1$. If $b_{n-1} = 0$, then $n-1$ *must* be placed before n. If $b_{n-1} = 1$, then $n-1$ *must* be placed after n.

$\vdots$

$n-k$: (*general step*) Consider b_{n-k}. We are given that $0 \le b_{n-k} \le k$. In steps n through $n-k+1$, the k numbers $n, n-1, \cdots, n-k+1$ have already been placed in the required order. If $b_{n-k} = 0$, then $n-k$ *must* be placed before all the numbers from step $n-k+1$. If $b_{n-k} = 1$, then $n-k$ *must* be placed between the first two numbers. ... If $b_{n-k} = k$, then $n-k$ *must* be placed after all the numbers.

$\vdots$

1: We *must* place 1 after the b_1st number in the sequence constructed in step $n-1$.

- Algorithm II

  We begin with n empty locations, labeling $1, 2, \ldots, n$ from left to right.

  1: Since there are to be b_1 integers that precede 1 in the permutation, we must put 1 in location number $b_1 + 1$.



Inversions in Permutations V

2: Since there are to be b_2 integers that precede 2 in the permutation, and since these integers have not yet been inserted, we must leave exactly b_2 locations for them. Thus, counting from the left, we put 2 in the $(b_2 + 1)$st empty location.

$\vdots$

k: (*general step*) Since there are to be b_k integers that precede k in the permutation, and since these integers have not yet been inserted, we must leave exactly b_k locations for them. Counting from the left, we put k in the $(b_k + 1)$st empty location.

$\vdots$

n: We put n in the one remaining empty location.



Inversions in Permutations VI

Example

Determine the permutation of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ whose inversion sequence is $5, 3, 4, 0, 2, 1, 1, 0$.

Algorithm I		Algorithm II								
			(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
8 :	8	1 :						1		
7 :	87	2 :				2		1		
6 :	867	3 :				2		1	3	
5 :	8657	4 :	4			2		1	3	
4 :	48657	5 :	4			2	5	1	3	
3 :	486537	6 :	4		6	2	5	1	3	
2 :	4862537	7 :	4		6	2	5	1	3	7
1 :	48625137	8 :	4	8	6	2	5	1	3	7



Inversions in Permutations VII

- The correspondence which associates the inversion sequence to each permutation is a one-to-one correspondence between the permutations of $\{1, 2, \ldots, n\}$ and the sequences of integers $b_1, b_2, \ldots, b_n$ satisfying

$$0 \leq b_1 \leq n-1,\ 0 \leq b_2 \leq n-2,\ \ldots,\ 0 \leq b_{n-1} \leq 1,\ b_n = 0.$$

- It is customary to call a permutation $i_1 i_2 \cdots i_n$ of $\{1, 2, \ldots, n\}$ *even* or *odd* according as its number of inversions is even or odd. The *sign* of the permutation is then defined to be $+1$ or -1 according to whether it is even or odd.



Inversions in Permutations VIII

Example

The determinant of an $n \times n$ matrix

$$A = [a_{ij}] \qquad (i, j = 1, 2, \ldots, n)$$

is defined to be

$$\det(A) = \sum \epsilon(i_1 i_2 \cdots i_n) a_{1 i_1} a_{2 i_2} \cdots a_{n i_n},$$

where the summation extends over all permutations $i_1 i_2 \cdots i_n$ of the set $\{1, 2, \ldots, n\}$, and $\epsilon(i_1 i_2 \cdots i_n)$ is equal to the sign of $i_1 i_2 \cdots i_n$.

Example

Bring the permutation 361245 to 123456 by successive switches of adjacent numbers.



Inversions in Permutations IX

Solution

Successive switches are as follows:

$$
\begin{array}{cccccc}
3 & 6 & 1 & 2 & 4 & 5 \\
3 & 1 & 6 & 2 & 4 & 5 \\
1 & 3 & 6 & 2 & 4 & 5 \\
1 & 3 & 2 & 6 & 4 & 5 \\
1 & 2 & 3 & 6 & 4 & 5 \\
1 & 2 & 3 & 4 & 6 & 5 \\
1 & 2 & 3 & 4 & 5 & 6
\end{array}
$$



Generating Combinations I

- Take S to be the set $S = \{x_{n-1}, \ldots, x_1, x_0\}$, we now see an algorithm that generates all of the 2^n combinations (subsets) of S.
- If we use 1 to denote that an element belongs and 0 to denote that an element does not belong, we can identify the 2^n combinations of S with the 2^n n-tuples

$$(a_{n-1}, \ldots, a_1, a_0) = a_{n-1} \cdots a_1 a_0$$

of 0's and 1's.

Example

$n = 3$.

	$\emptyset$	$\{x_0\}$	$\{x_1\}$	$\{x_1, x_0\}$	$\{x_2\}$	$\{x_2, x_0\}$	$\{x_2, x_1\}$	$\{x_2, x_1, x_0\}$
a_2	0	0	0	0	1	1	1	1
a_1	0	0	1	1	0	0	1	1
a_0	0	1	0	1	0	1	0	1

Generating Combinations II

Example

Let $S = \{x_6, x_5, x_4, x_3, x_2, x_1, x_0\}$. The 7-tuple corresponding to the combination $\{x_5, x_4, x_2, x_0\}$ is 0110101. The combination corresponding to the 7-tuple 1010001 is $\{x_6, x_4, x_0\}$.

Example

Let $n = 7$. The number 29 is between 0 and $2^7 - 1 = 127$ and can be expressed as

$$29 = 0 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0.$$

Thus, 29 has a binary numeral of 7 digits given by 0011101 and corresponds to the combination $\{x_4, x_3, x_2, x_0\}$ of the set $S = \{x_6, x_5, x_4, x_3, x_2, x_1, x_0\}$.

We write down the numbers from 0 to $2^n - 1$ in increasing order by size, but in binary form, adding 1 each time, using base 2 arithmetic!



Generating Combinations III

Example

Generate the 4-tuples of 0's and 1's.

Number	0	1	2	3	4	5	6	7
Binary Numeral	0000	0001	0010	0011	0100	0101	0110	0111
	8	9	10	11	12	13	14	15
	1000	1001	1010	1011	1100	1101	1110	1111

Example

Using the scheme just described, what is the combination of $\{x_6, x_5, x_4, x_3, x_2, x_1, x_0\}$ that immediately follows the combination $\{x_6, x_4, x_2, x_1, x_0\}$?

Solution

The combination $\{x_6, x_4, x_2, x_1, x_0\}$ corresponds to the binary numeral 1010111, followed immediately by 1011000, which corresponds to the combination $\{x_6, x_4, x_3\}$.

Generating Combinations IV

Example

Which combination of $S = \{x_6, x_5, x_4, x_3, x_2, x_1, x_0\}$ is 108th on the list?

Solution

The base 2 numeral for 108
$(= 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 0 \times 2^0)$ is 1101100 so that it corresponds to the combination $\{x_6, x_5, x_3, x_2\}$.



Generating Combinations V

Base 2 algorithm for generating the combinations of $\{x_{n-1}, \ldots, x_1, x_0\}$

Begin with $a_{n-1} \cdots a_1 a_0 = 0 \cdots 00$. While $a_{n-1} \cdots a_1 a_0 \neq 1 \cdots 11$, do the following:

1. Find the smallest integer j (between $n-1$ and 0) such that $a_j = 0$.
2. Replace a_j by 1 and each of $a_{j-1}, \ldots, a_0$ (which by our choice of j all equal 1) by 0.

The algorithm comes to an end when $a_{n-1} \cdots a_1 a_0 = 1 \cdots 11$, which is the last binary n-tuple on the resulting list.

- The rule of succession given in the algorithm is a consequence of addition using base 2 arithmetic. This scheme is called the *lexicographic ordering of n-tuples*. It sometimes called the *squashed ordering of combinations*.



Generating Combinations VI

- Is it possible to generate the combinations of a set of n elements in a different order so that the immediate successor of a combination differs from it as little as possible?

Example

$n = 1, 2, 3$.

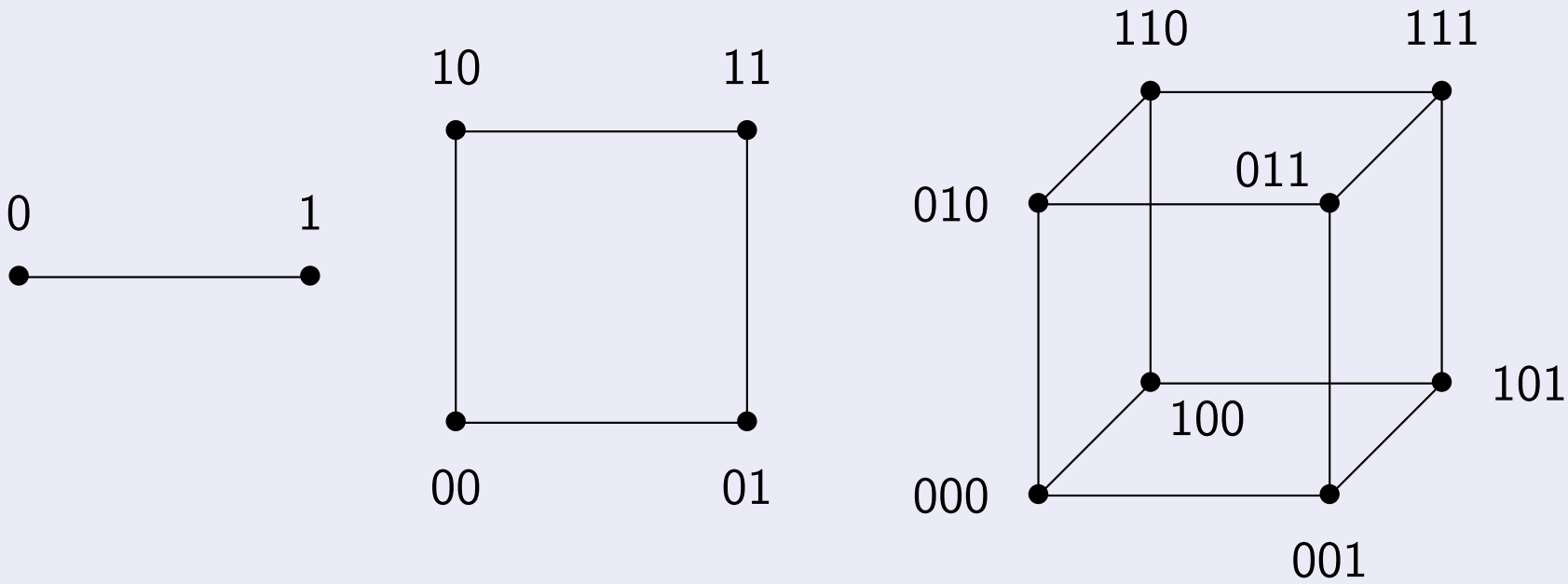
$n=1$		$n=2$		$n=3$	
$\emptyset$	0	$\emptyset$	0 0	$\emptyset$	0 0 0
$\{x_0\}$	1	$\{x_0\}$	0 1	$\{x_0\}$	0 0 1
		$\{x_1, x_0\}$	1 1	$\{x_1, x_0\}$	0 1 1
		$\{x_1\}$	1 0	$\{x_1\}$	0 1 0
				$\{x_2, x_1\}$	1 1 0
				$\{x_2, x_1, x_0\}$	1 1 1
				$\{x_2, x_0\}$	1 0 1
				$\{x_2\}$	1 0 0



Generating Combinations VII

Example

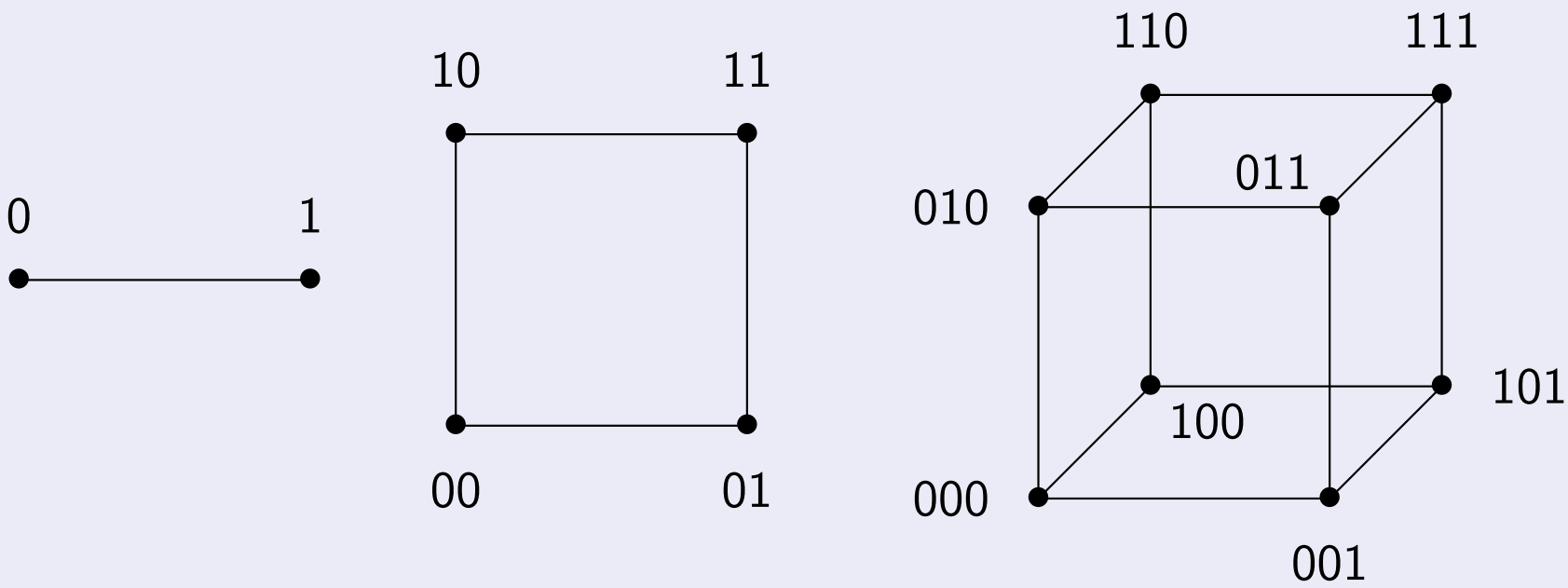
Geometric identification – $n = 1, 2, 3$.



Generating Combinations VIII

Example

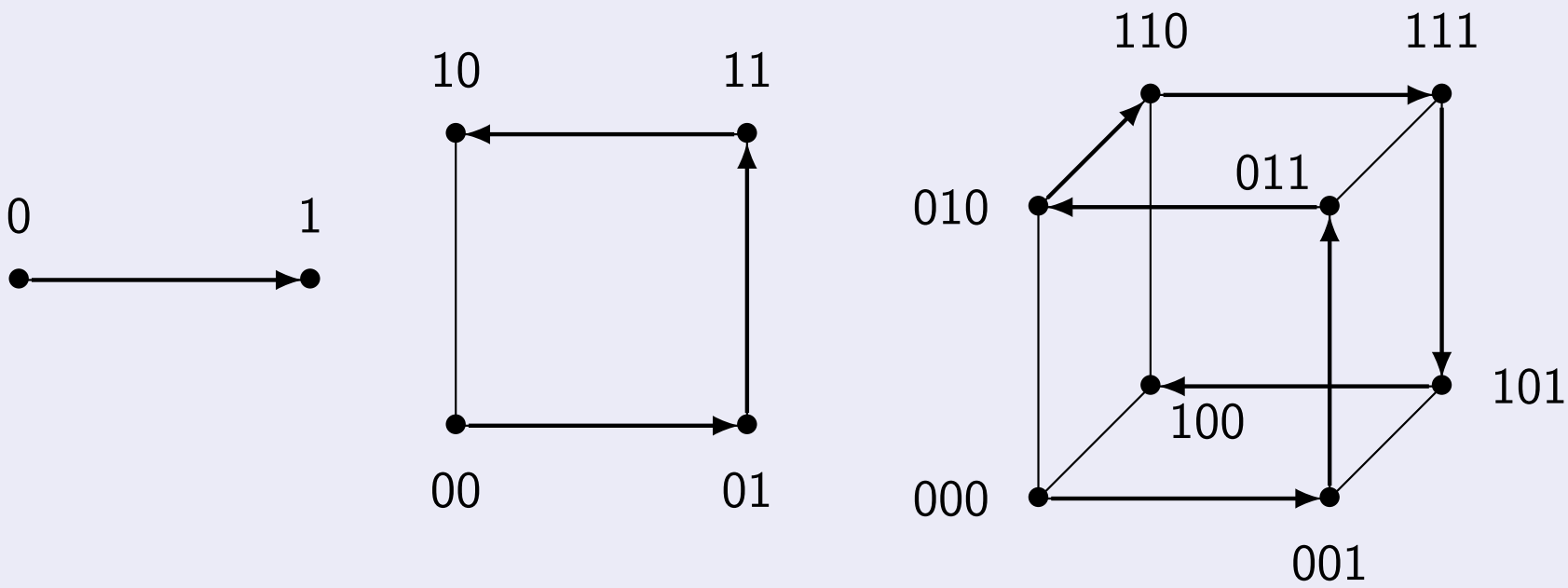
Gray code and *cyclic* Gray code – *Walks* and *cyclic* walks.



Generating Combinations VIII

Example

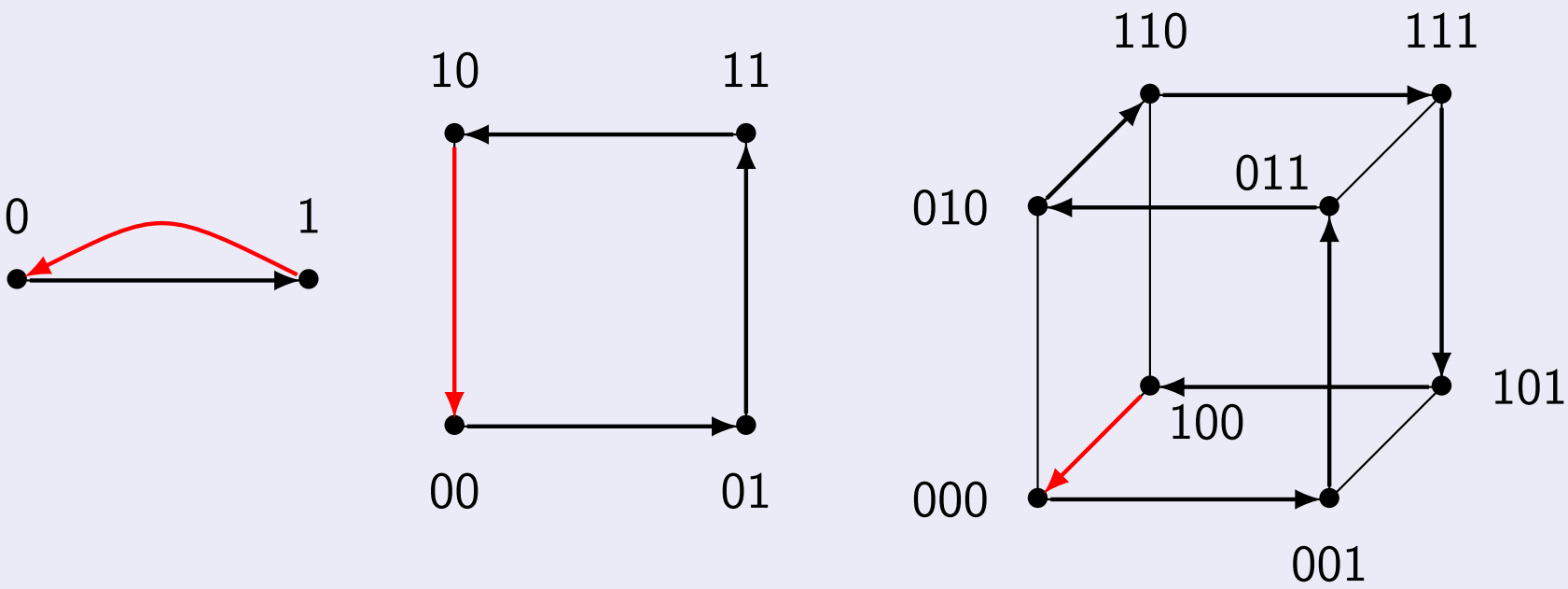
Gray code and *cyclic* Gray code – *Walks* and *cyclic* walks.



Generating Combinations VIII

Example

Gray code and *cyclic* Gray code – *Walks* and *cyclic* walks.



Generating Combinations IX

- The general inductive definition of the *reflected Gray code* of order n is the following:
  1. The reflected Gray code of order 1 is $\begin{matrix} 0 \\ 1 \end{matrix}$.
  2. Suppose $n > 1$ and the reflected Gray code of order $n-1$ has been constructed. To construct the reflected Gray code of order n, we first list the $(n-1)$-tuples of 0's and 1's in the order given by the reflected Gray code of order $n-1$, and attach a 0 at the beginning of each $(n-1)$-tuple. We then list the $(n-1)$-tuples in the order which is the reverse of that given by the reflected Gray code of order $n-1$, and attach a 1 at the beginning.



Generating Combinations IX

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Generating Combinations IX

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Generating Combinations X

Algorithm for generating the n-tuples of 0's and 1's in the reflected Gray code order

Begin with the tuples $a_{n-1}a_{n-2}\cdots a_0 = 00\cdots 0$. While the n-tuples $a_{n-1}a_{n-2}\cdots a_0 \neq 10\cdots 0$, do the following:

1. Compute $\sigma(a_{n-1}a_{n-2}\cdots a_0) = a_{n-1} + a_{n-2} + \cdots + a_0$.
2. If $\sigma(a_{n-1}a_{n-2}\cdots a_0)$ is even, change a_0 (from 0 to 1 or 1 to 0).
3. Else, determine j such that $a_j = 1$ and $a_i = 0$ for all i which $j > i$, and then change a_{j+1} (from 0 to 1 or 1 to 0).



Generating Combinations X

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Generating Combinations X

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Generating Combinations X

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2. If $\sigma(a_{n-1}a_{n-2}\cdots a_0)$ is even, change a_0 (from 0 to 1 or 1 to 0).
3. Else, determine j such that $a_j = 1$ and $a_i = 0$ for all i which $j > i$, and then change a_{j+1} (from 0 to 1 or 1 to 0).



Generating Combinations XI

Example

Gray code for $n = 4$

0	0	0	0
0	0	0	1
0	0	1	1
0	0	1	0
0	1	1	0
0	1	1	1
0	1	0	1
0	1	0	0
1	1	0	0
1	1	0	1
1	1	1	1
1	1	1	0
1	0	1	0
1	0	1	1
1	0	0	1
1	0	0	0



Generating Combinations XII

Theorem

The algorithm for generating the n-tuples of 0's and 1's previously described produces the reflected Gray code of order n for each positive integer n.

Proof

It is clear that the algorithm applied to $n = 1$ produces the reflected Gray code of order 1. Let $n > 1$ and assume that the algorithm applied to $n - 1$ produces the reflected Gray code of order $n - 1$. The first 2^{n-1} n-tuples of the reflected Gray code of order n consist of the $(n - 1)$-tuples of the reflected Gray code of order $n - 1$. with a 0 attached at the beginning of each $(n - 1)$-tuple. Since the $(n - 1)$-tuple $10 \cdots 0$ occurs last in the reflected Gray code of order $n - 1$



Generating Combinations XII

Theorem

The algorithm for generating the n-tuples of 0's and 1's previously described produces the reflected Gray code of order n for each positive integer n.

Proof

It is clear that the algorithm applied to $n = 1$ produces the reflected Gray code of order 1. Let $n > 1$ and assume that the algorithm applied to $n - 1$ produces the reflected Gray code of order $n - 1$. The first 2^{n-1} n-tuples of the reflected Gray code of order n consist of the $(n - 1)$-tuples of the reflected Gray code of order $n - 1$. with a 0 attached at the beginning of each $(n - 1)$-tuple. Since the $(n - 1)$-tuple $10 \cdots 0$ occurs last in the reflected Gray code of order $n - 1$



Generating Combinations XIII

Example

Determine the 8-tuples that are successors of 10100110, 00011111, and 01010100 in the reflected Gray code of order 8.

Solution

10100111, 00011101, and 01011100.



Generating Combinations XIII

Example

Determine the 8-tuples that are successors of 10100110, 00011111, and 01010100 in the reflected Gray code of order 8.

Solution

10100111, 00011101, and 01011100.



Generating r-Combinations I

Example

In the previous section we listed all the 4-combinations of $\{1, 2, 3, 4\}$ in the squashed ordering. Selecting the 2-combinations from among them we get the squashed ordering of the 2-combinations of $\{1, 2, 3, 4\}$:

$$\{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\}, \quad \{1, 4\}, \quad \{2, 4\}, \quad \{3, 4\}.$$

All the 4-combinations are

$\emptyset$	$\{1\}$	$\{2\}$	$\{1, 2\}$
$\{3\}$	$\{1, 3\}$	$\{2, 3\}$	$\{1, 2, 3\}$
$\{4\}$	$\{1, 4\}$	$\{2, 4\}$	$\{1, 2, 4\}$
$\{3, 4\}$	$\{1, 3, 4\}$	$\{2, 3, 4\}$	$\{1, 2, 3, 4\}$



Generating r-Combinations II

- Let A and B be two r-combinations of the set $S = \{1, 2, \cdots, n\}$. Then we say that A *precedes* B *in the lexicographic order* provided that the smallest integer which is in their union $A \cup B$, but not in their intersection $A \cap B$ (that is, in one but not both of the sets), is in A.

Example

Let 5-combinations A and B of $\{1, 2, 3, 4, 5, 6, 7, 8\}$ be given by $A = \{2, 3, 4, 7, 8\}$ and $B = \{2, 3, 5, 6, 7\}$. The smallest element that is in one, but not both, of the sets is 4 (which belongs to A). Hence A precedes B in the lexicographic order.



Generating r-Combinations III

- Let us agree that, whenever we write a combination of $\{1, 2, \cdots, n\}$, we write the integers in it from smallest to largest. Thus, we agree that an r-combination of $S = \{1, 2, \cdots, n\}$ is to be written in the form

$$a_1, a_2, \cdots, a_r, \quad \text{where } 1 \leq a_1 < a_2 < \cdots < a_r \leq n.$$

Let us also agree, for convenience, to write this r-combination as

$$a_1 a_2 \cdots a_r$$

without commas, that is, as a word of length r.

Example

In the previous example, we write $A = 23478$ and $B = 23567$. We see that A and B agree in their first two letters and disagree in their third letter. Since $4 < 5$ (4 comes earlier in our alphabet than 5), A precedes B in the lexicographic order.



Generating r-Combinations IV

Example

We consider the lexicographic order of the 5-combinations of $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. The first 5-combination is clearly 12345; the last 5-combination is 56789. What 5-combination immediately follows 12389 (in our dictionary)? Among the 5-combinations that begin with 123, 12389 is the last. Among the 5-combinations that begin with 12 and don't have a 3 in the third position, 12456 is the first. Thus, 12456 immediately follows 12389.



Generating r-Combinations V

Theorem

Let $a_1 a_2 \cdots a_r$ be an r-combination of $\{1, 2, \ldots, n\}$. The first r-combination in the lexicographic ordering is $12 \cdots r$. The last r-combination in the lexicographic ordering is $(n-r+1)(n-r+2) \cdots n$. Assume that $a_1 a_2 \cdots a_r \neq (n-r+1)(n-r+2) \cdots n$. Let k be the largest integer such that $a_k < n$ and $a_k + 1$ is different from each of $a_1 a_2 \cdots a_r$. Then the r-combination that is the immediate successor of $a_1 a_2 \cdots a_r$ in the lexicographic ordering is

$$a_1 \cdots a_{k-1}(a_k+1)(a_k+2) \cdots (a_k+r-k+1).$$



Generating r-Combinations VI

Algorithm for generating the r-combinations of $\{1, 2, \ldots, n\}$ in lexicographic order

Begin with the r-combination $a_1 a_2 \cdots a_r = 12 \cdots r$. While $a_1 a_2 \cdots a_r \neq (n-r+1)(n-r+2) \cdots n$, do the following:

1. Determine the largest integer k such that $a_k + 1 \leq n$ and $a_k + 1$ is not one of $a_1, a_2, \ldots, a_r$.
2. Replace $a_1 a_2 \cdots a_r$ with the r-combination

$$a_1 \cdots a_{k-1}(a_k + 1)(a_k + 2) \cdots (a_k + r - k + 1).$$



Generating r-Combinations VII

Example

Generate the 4-combinations of $\{1, 2, 3, 4, 5, 6\}$.

1234	1235	1236	1245	1246
1256	1345	1346	1356	1456
2345	2346	2356	2456	3456.

Example

Generate the 3-permutations of $\{1, 2, 3, 4\}$. We first generate the 3-combinations in lexicographic order: 123, 124, 134, 234. For each 3-combination, we then generate all of its permutations:

123	132	312	321	231	213
124	142	412	421	241	214
134	143	413	431	341	314
234	243	423	432	342	324.

Generating r-Combinations VIII

Theorem

The r-combination $a_1 a_2 \cdots a_r$ of $\{1, 2, \ldots, n\}$ occurs in place number

$$\binom{n}{r} - \binom{n-a_1}{r} - \binom{n-a_2}{r-1} - \cdots - \binom{n-a_{r-1}}{2} - \binom{n-a_r}{1}$$

in the lexicographic order of the r-combinations of $\{1, 2, \cdots, n\}$.



Generating r-Combinations IX

Proof

We first count the number of r-combinations that come after $a_1 a_2 \cdots a_r$

1. There are $\binom{n-a_1}{r}$ r-combinations whose first element is greater than a_1 that come after $a_1 a_2 \cdots a_r$.
2. There are $\binom{n-a_2}{r-1}$ r-combinations whose first element is a_1 but whose second element is greater than a_2 that come after $a_1 a_2 \cdots a_r$.

$\vdots$

$r-1$. There are $\binom{n-a_{r-1}}{2}$ r-combinations that begin $a_1 \cdots a_{r-2}$ but whose $(r-1)$st element is greater than a_{r-1} that come after $a_1 a_2 \cdots a_r$.

r. There are $\binom{n-a_r}{1}$ r-combinations that begin $a_1 \cdots a_{r-1}$ but whose rth element is greater than a_r that come after $a_1 a_2 \cdots a_r$.

Subtracting the number of r-combinations that come after $a_1 a_2 \cdots a_r$ from the total number $\binom{n}{r}$ of r-combinations, we find that the place of $a_1 a_2 \cdots a_r$ is as given in the theorem.



Partial Orders and Equivalence Relations I

- Orders are "*total* orders", in the sense that there is a first object, a second object, a third object, ..., a last object.
- Orders are "*partial* orders", in the sense that, given any two sets, neither may be a subset of the other, and given any two integers, neither may be divisible by the other.

Example

Let $X = \{1, 2, 3, 4, 5, 6\}$. Write $a|b$ to mean that a is a divisor of b (equivalently, b is divisible by a). This defines a partial order on X and we have, for example, $2|6$ and $3 \not| 5$.

- Let X be a set. A *relation* R on X is a "property" that may or may not hold between any two given elements of X. More formally, a relation on X is a subset R of the set $X \times X$ of ordered pairs of elements of X. We write aRb, provided that the ordered pair (a, b) belongs to R; we also write $a \not R b$ whenever (a, b) is not in R.
- The following are special properties that a relation R on a set X may have:



Partial Orders and Equivalence Relations II

1. R is *reflexive*, provided that xRx for all x in X.
2. R is *irreflexive*, provided that $x \not R x$ for all x in X.
3. R is *symmetric*, provided that, for all x and y in X, whenever we have xRy we also have yRx.
4. R is *antisymmetric*, provided that, for all x and y in X with $x \neq y$, whenever we have xRy, we also have $y \not R x$. Equivalently, for all x and y in X, xRy and yRx together imply that $x = y$.
5. R is *transitive*, provided that, for all x, y, z in X, whenever we have xRy and yRz, we also have xRz.

Example

Relation	reflexive	irreflexive	transitive	antisymmetric
$\subseteq$	yes	no	yes	yes
$\mid$	yes	no	yes	yes (for positive integers)
$\subset$	no	yes	yes	yes
$\leq$	yes	no	yes	yes
$<$	no	yes	yes	yes

Partial Orders and Equivalence Relations III

- A *partial order* on a set X is a reflexive, antisymmetric, and transitive relation R. A *strict partial order* on a set X is an irreflexive, antisymmetric, and transition relation.

Example

$\subseteq$, $\leq$, and $|$ are partial orders, while $\subset$ and $<$ are strict partial orders.

- A set X on which a partial order $\leq$ is defined is sometimes referred to as a *partially ordered set* (or sometimes simply as a *poset*) and denoted by $(X, \leq)$.
- If R is a relation on a set X, then for x and y in X, x and y are *comparable*, provided that either xRy or yRx; x and y are *incomparable* otherwise.



Partial Orders and Equivalence Relations IV

- A partial order R on a set X is a *total order*, provided that every pair of elements of X is comparable. The standard relation $\leq$ on a set of numbers is a total order.

Theorem

Let X be a finite set with n elements. Then there is a one-to-one correspondence between the total orders on X and the permutations of X. In particular, the number of different total orders on X is $n!$.

1. We show by induction on n that each total order $\leq$ on X corresponds to a permutation $a_1, a_2, \ldots, a_n$ of X with $a_1 < a_2 < \cdots < a_n$.
2. If $n = 1$, this is trivial. Let $n > 1$. We first show that there is a *minimal element* of X, that is, an element a_1 such that $b \leq a_1$ implies $b = a_1$ (equivalently, there is no element x with $x < a_1$).
3. Let a be any element of X. If a is not the minimal element, then there is an element b such that $b < a$. If b is not a minimal element, there is an element c such that $c < b$ so that $c < b < a$.



Partial Orders and Equivalence Relations V

4. Continuing like this and using the fact that X is a finite set, eventually we locate a minimal element a_1.
5. Suppose there is an element $x \neq a_1$ of X such that $a_1 \not< x$.
6. Since we have a total order, we must have $x < a_1$ contradicting the minimality of a_1. Hence $a_1 < x$ for all x in X different from a_1.
7. Applying induction to the set of $n-1$ elements of X different from a_1, we conclude that these elements can be ordered $a_2, a_3, \ldots, a_n$ with $a_2 < a_3 < \cdots < a_n$.

Hence $a_1, a_2, \ldots, a_n$ is a permutation of the elements of X with $a_1 < a_2 < \cdots < a_n$. □

- As a result, a finite totally ordered set is often denoted as $a_1 < a_2 < \cdots < a_n$, or simply as a permutation $a_1, a_2, \ldots, a_n$.
- Let a and b be in X. Then a is *covered by* b (also expressed as b *covers* a), denoted $a <_c b$, provided that $a < b$ and no elements c can be squeezed between a and b; that is, there is no element c such that both $a < c$ and $c < b$ hold.



Partial Orders and Equivalence Relations VI

If $(X, \leq)$ is a totally ordered set, then the elements of X can be listed as $x_1, x_2, \ldots, x_n$ such that $x_1 <_c x_2 <_c \cdots <_c x_n$. It is for this reason that a totally ordered set is also called a *linearly ordered set*.

Example

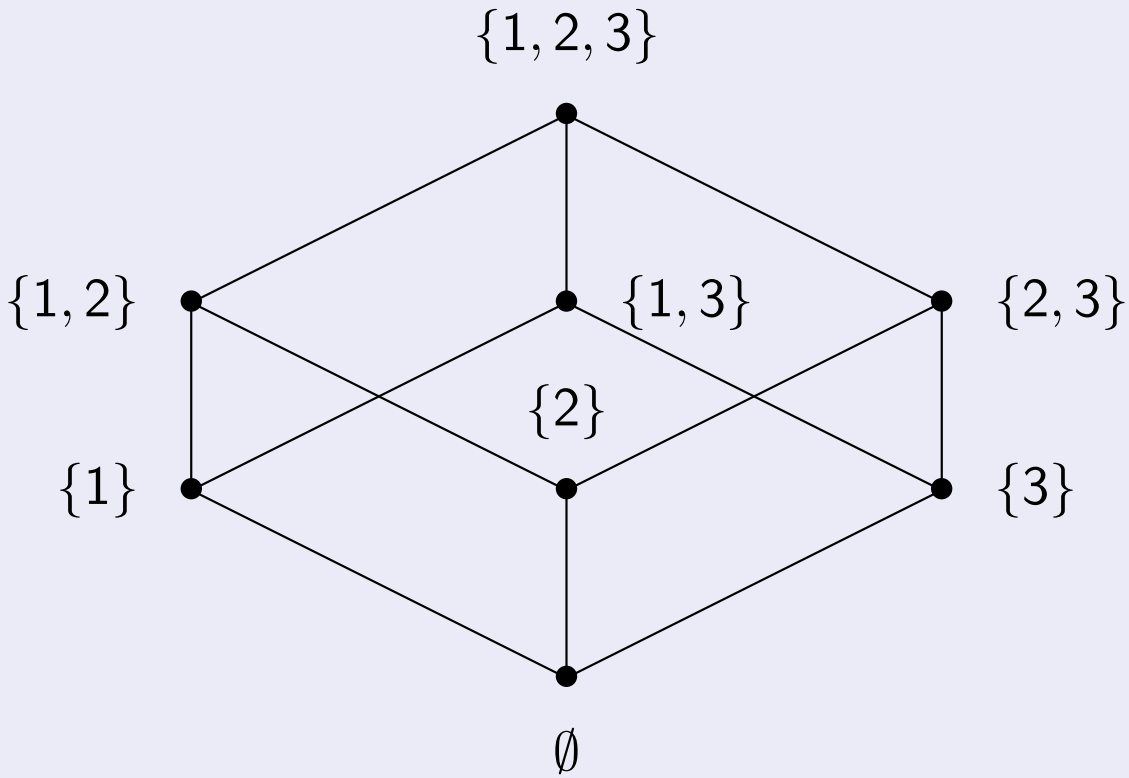
A totally ordered set of 5 elements is represented by the diagram



Partial Orders and Equivalence Relations VII

Example

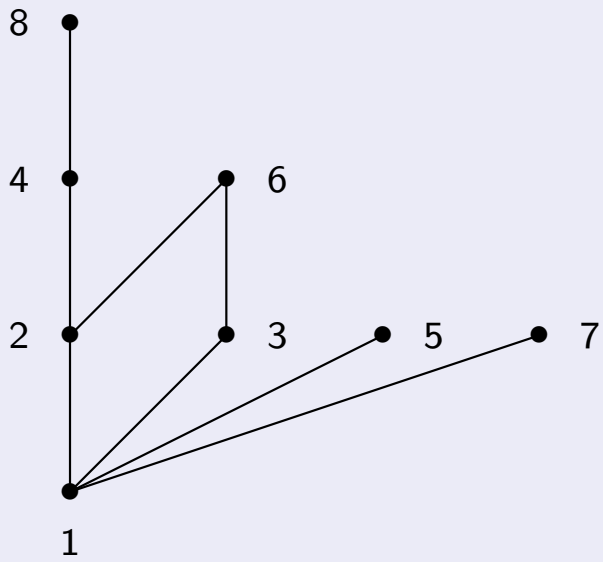
The partially ordered set of subsets of the set $\{1, 2, 3\}$ is represented by the diagram



Partial Orders and Equivalence Relations VIII

Example

The set of the first eight positive integers, partially ordered by "is a divisor of", is represented by the diagram



Partial Orders and Equivalence Relations IX

- Let $\leq_1$ and $\leq_2$ be two partial orders on the same set X. Then the partially ordered set $(X, \leq_2)$ is an *extension* of the partially ordered set $(X, \leq_1)$, provided that whenever $a \leq_1 b$ holds, $a \leq_2 b$ also holds. In particular, an extension of a partially ordered set has more comparable pairs.

Theorem

Let $(X, \leq)$ be a finite partially ordered set. Then there is a linear extension of $(X, \leq)$.



Partial Orders and Equivalence Relations X

Algorithm for a linear extension of a partially ordered set

1. Choose a minimal element x_1 of X (with respect to the partial order $\leq$).
2. Delete x_1 from X and choose a minimal element x_2 from among the remaining $n-1$ elements.
3. Delete, in addition, x_2 from X, and choose a minimal element x_3 from among the remaining $n-2$ elements.

   $\vdots$
4. Delete x_{n-1} from X, leaving exactly one element x_n.

Example

Let $X = \{1, 2, \cdots, n\}$ be the set consisting of the first n positive integers, and consider the partially ordered set $(X, |)$, where, as before, $|$ means "is a divisor of". Since, if $i|j$, then i is smaller than j, it follows that $1, 2, \cdots, n$ is a linear extension of $(X, \leq)$.



Partial Orders and Equivalence Relations XI

Example

Let X be a set of n elements and consider the partially ordered set $(\mathcal{P}(X), \subseteq)$ of all subsets of X partially ordered by containment. Since, if $A \subseteq B$ implies that $|A| \leq |B|$, it follows that, if we start with the empty set, list all the one-element subsets in some order, then the two-element subsets in some order, then the three-element subsets in some order, and so on, we obtain a linear extension of $(\mathcal{P}(X), \subseteq)$. For instance in $n = 3$ and $X = \{1, 2, 3\}$, then

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$$

is a linear extension of $(\mathcal{P}(X), \subseteq)$.

- Let X be a set. A relation R on X is an *equivalence relation*, provided it is reflective, symmetric, and transitive. A relation that is an equivalence relation is usually denoted by $\sim$.



Partial Orders and Equivalence Relations XII

- Let $\sim$ be an equivalence relation on X. For each a in X, the *equivalence class* of a is the set

$$[a] = \{x : x \sim a\}$$

of all elements equivalent to a.

Theorem

Let $\sim$ be an equivalence relation on a set X. Then the distinct equivalence classes partition X into nonempty parts. Conversely, given any partition of X into nonempty parts, there is an equivalence relation on X whose equivalence classes are the parts of the partition.

